

## OBJECTIVE WORKSHEET

# Heat

Science — Physics · Class 7 · Max Marks: 25 · Time: 45 min

**Section A — Multiple Choice Questions ( $1 \times 8 = 8$ )****1. The unit in which temperature is usually measured on a thermometer is the:**

- A. metre
- B. kilogram
- C. second
- D. degree Celsius ( $^{\circ}\text{C}$ )

**2. The normal temperature of a healthy human body is about:**

- A.  $35^{\circ}\text{C}$
- B.  $37^{\circ}\text{C}$
- C.  $100^{\circ}\text{C}$
- D.  $42^{\circ}\text{C}$

**3. The temperature range of a clinical thermometer is about:**

- A.  $35^{\circ}\text{C}$  to  $42^{\circ}\text{C}$
- B.  $-10^{\circ}\text{C}$  to  $110^{\circ}\text{C}$
- C.  $0^{\circ}\text{C}$  to  $250^{\circ}\text{C}$
- D.  $0^{\circ}\text{C}$  to  $100^{\circ}\text{C}$

**4. Heat always flows on its own from a body at \_\_\_\_ temperature to a body at \_\_\_\_ temperature.**

- A. zero, high
- B. lower, higher
- C. higher, lower
- D. equal, equal

**5. Which of the following is the best conductor of heat?**

- A. plastic
- B. wood
- C. rubber
- D. aluminium

**6. The other end of a steel spoon kept in a cup of hot tea becomes hot by the process of:**

- A. conduction
- B. radiation
- C. evaporation
- D. convection

**7. A sea breeze blows from the sea towards the land during the:**

- A. midnight
- B. night
- C. day
- D. early hours

**8. The heat of the Sun reaches the Earth through empty space mainly by:**

- A. conduction
- B. radiation
- C. all three
- D. convection

**Section B — Fill in the Blanks ( $1 \times 5 = 5$ )**

1. The device used to measure temperature is a \_\_\_\_\_.
2. A clinical thermometer has a \_\_\_\_\_ near the bulb that stops the mercury falling on its own.
3. Materials that do not allow heat to pass through them easily are called \_\_\_\_\_.
4. In \_\_\_\_\_, the particles of the medium do not move from their place while passing on heat.
5. Dark-coloured surfaces are \_\_\_\_\_ absorbers of heat than light-coloured surfaces.

**Section C — True or False ( $1 \times 5 = 5$ )**

1. Our sense of touch is a reliable way to measure the exact temperature of an object. ( True / False )
2. Water is a poor conductor of heat. ( True / False )
3. Convection can take place in solids. ( True / False )
4. Woollen clothes keep us warm because the trapped air is a poor conductor of heat. ( True / False )
5. A laboratory thermometer is used to measure the temperature of the human body. ( True / False )

#### Section D — Match the Columns (1 × 4 = 4)

| Column A                | Column B                                    |
|-------------------------|---------------------------------------------|
| 1. Clinical thermometer | a) a good conductor of heat                 |
| 2. Land breeze          | b) reduces heat transfer by all three modes |
| 3. Copper               | c) measures body temperature                |
| 4. Thermos flask        | d) blows from land to sea at night          |

#### Section E — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** We wear light-coloured clothes in summer.  
**Reason (R):** Light-coloured surfaces reflect most of the heat radiation falling on them.
2. **Assertion (A):** The handles of cooking pans are made of plastic or wood.  
**Reason (R):** Plastic and wood are good conductors of heat.
3. **Assertion (A):** A sea breeze blows during the day.  
**Reason (R):** During the day the land heats up faster than the sea, so the warm air over the land rises.

### Answer Key

#### Multiple Choice Questions

- |      |      |      |      |      |
|------|------|------|------|------|
| 1. D | 2. B | 3. A | 4. C | 5. D |
| 6. A | 7. C | 8. B |      |      |

#### Fill in the Blanks

1. thermometer
2. kink (constriction)
3. insulators (poor conductors)
4. conduction
5. better (good)

#### True or False

1. False
2. True
3. False
4. True
5. False

#### Match the Columns

1-c, 2-d, 3-a, 4-b

#### Assertion–Reason

1. A

2. C

3. A

